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Case Study

A Rare Atypical Mycobacterial Infection Following Face Lifting Thread Implantation: A Case Study in New Delhi, India

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Abstract

Background: Infection is a common complication after the thread-lifting procedure, but mycobacterial infection after thread-lifting procedure is quite rare. Objective: To study the prevalence of adverse events during eight weeks after thread implant. Patients/Methods: A 63-year-old woman undergone face lift definisse thread implantation at cosmetic studio in India. After implantation, patient felt boils on her face and immediately consulted doctor. A surgical pathology report of patient confirmed atypical mycobacterial infection as clinical diagnosis. Consequently, patient is under antibiotics treatment. Result: Facial rejuvenation techniques with thread lifting have gained popularity. Atypical mycobacterial infection with the thread lifting procedure is considered as rare adverse event. Currently, the patient is in recovering stage. Conclusion: Face-lifting surgery is not completely safe procedure for the patients. Presented case study with the experience of the patient should be taken as a guide to prevent mycobacterial infection after face lifting thread implant in future.

Keywords: Definisse thread, mycobacterial infection, adverse event, surgery.

Introduction

A slender, smooth, wrinkle-free face is a desirable feature associated with youth and beauty. It is inevitable that the proportion of collagen fibers and fibroblasts located in the skin dermis gradually decreases as people age [1]. Aging is a natural process that involves factors of intrinsic (genetic and epigenetic) and external origin (nutrition, radiation, temperature and stress) and gradually deteriorates the organs/organ systems in an organism [2]. The aging phenomenon affects the skin in reduction in fat volume and bone support, sagging of the skin, reduction in the skin's elasticity and thickness and the degeneration of the soft tissue supporting ligaments. Surgeons offer surgical and non-surgical procedures to rejuvenate the aging of the face and neck. Current approaches to face and

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neck rejuvenation include laser treatment, injections with fillers, calcium hydroxylapatite, thread-lifting, intensity focused ultrasound (IFU) and incobotulinumtoxin injections [2]. Thread lifting is a minimally invasive aesthetic procedure that lifts and realigns sagging tissue [3]. The DefinisseTM (also known as Happy LiftTM) thread lift treatment uses barbed absorbable threads to suspend tissues and induce biostimulation [4]. The Definisse Thread treatment is a double-needle, absorbable, monofilament, suspension, barbed thread composed of synthetic poly-L-lactic acid and caprolactone polymers. The biostimulatory action of the thread induces fibrosis and facilitates tissue suspension, which in turn leads to a lifting effect for at least 12 months before the thread is completely resorbed in the body [4]. Currently, global market is also moving towards thread lifting with completely absorbable suspension technology for suspending facial tissue and restoring facial volume. In this article, we described a case of definisse face-lift thread causing atypical mycobacterial infection in the patient.

Case Report

A 63-year-old Indian woman undergone a face-lift definise thread implantation procedure at a cosmetic studio in India. After 10 days of implantation, the patient felt boils on her face and immediately consulted a medical doctor. The medical doctor suggested patient to take antibiotics- amoxycillin (1000 mg), Ceftazidime (500 mg) & Ciprofloxacin (500 mg) for 60 to 90 days but there was persistent pain at implantation site. After 18 days of implantation, the patient was advised to visit doctor's clinic, where the doctor partially removed the thread from the implantation site. Unfortunately, this procedure did not show any improvement in the patient´s condition, and the doctor did not provide any further course of treatment to the patient. The patient was experiencing an increase in pain, boils and bruises (Figure 1) at the implantation site. After 37 days of implantation, the patient was once again advised to visit doctor's clinic. During this visit, the patient requested the doctor for the complete removal of the implanted thread, but doctor refused to do so. Subsequently, after 51 days of implantation, the patient took a second consultation from another doctor in India. During this consultation, the patient was diagnosed with post thread granuloma, and the doctor prescribed medications and advised for a follow-up visit after 14 days. The patient did not feel any improvement, so next day she took a third consultation from another doctor in India. According to this doctor, the condition was identified as a bacterial infection, and a longer duration of treatment was required to heal the infection with antibiotics. To confirm the bacterial infection, the patient took a consultation from two additional doctors in Chandigarh & Himanchal Pradesh in India. Subsequently, the patient was diagnosed with bacterial/atypical mycobacterial infection. After viewing all the reports, the patient finally reached to Postgraduate Institute of Medical Education & Research (PGIMER) in Chandigarh after 1 month and 27 days of the implantation. The doctor advised the patient to undergo for a skin biopsy for Atypical mycobacterial culture test & histopathological examination. A skin biopsy sample of size 0.4×0.2×0.2 cm was tested in different labs of PGIMER such as clinical bacteriology, mycobacteriology and surgical pathology.



Figure 1. Formation of boils and bruises at implantation

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A report was generated from Clinical Bacteriology Lab, Department of Medical Microbiology at PGIMER, Chandigarh that showed the culture report as "Aerobic and Anaerobic Bacterial Culture Sterile after 48 hours of Incubation". Similarly, a report was issued from Mycobacteriology, Department of Medical Microbiology at PGIMER that showed the polymerase chain reaction (PCR) test for mycobacteria tuberculosis was Negative. A surgical pathology report was generated by the Department of Histopathology of PGIMER that showed the "Atypical mycobacterial infection" as clinical diagnosis. The examination of the skin of the face revealed features consistent with granulomatous inflammation- atypical mycobacterial infection. Skin biopsy examined showed epidermis and dermis. Epidermis showed spongiosis. The dermis showed granulation tissue and neutrophilic abscesses. Many epithelial histiocytic collections were seen forming ill formed granulomas in papillary and deep dermis. Fite stain for lepra was negative. The Ziehl–Neelsen (ZN) stain for Acid-Fast Bacilli (AFB) and the modified ZN stain both were negative. Grams and Periodic Acid-Schiff (PAS) stain did not highlight the presence of bacterial colonies or a fungal profile. Overall, these features were consistent with granulomatous inflammation consistent with atypical mycobacterial infection. Now, the patient is under antibiotics treatment Clarithromycin (500 mg) and Rifampicin (300 mg) for 90 days. Currently, patient is in recovering stage (Figure 2).



Figure 2 Post treatment view of implantation site.

Discussion

Atypical mycobacteria or nontuberculous mycobacteria are organisms that cause various diseases such as pulmonary infections, skin and soft tissue infections, disseminated infections, as well as a wide range of more rarely encountered infections [5]. These organisms are mostly found in soil, indoor & outdoor water sources, and are recognized to colonize poorly sanitized medical equipment. Atypical mycobacteria most commonly infect young children, immunocompromised individuals, individuals with indwelling medical equipments, and those who have recently undergone surgical or non-surgical procedures [6]. In the American Society for Dermatologic Surgery, there were two case reports of mycobacterial infections after thread lift procedures published in year 2007 & 2016. One case focused on an infection with Mycobacterium Fortuitum after the use of Polypropylene threads for face lifting was published by Christian Cruz Hofling in 2007 and the second case focused on the mycobacterium massiliense infection after thread lift insertion was published in 2016 by Jung Jin Shin [7-8]. Recently, a case study on Facial thread Lifting complications in China from April 2014 to January 2020 was published by Yi-Lin Li in The American Society of Plastic Surgeons. In this study, the patients with postoperative complications of thread face lift were treated after thread implantation. It was observed that 8.9 % of patients experienced either local infections or infections associated with the thread [9]. Infections related to face lifting threads are not reported in the international databases but the same case is also reported in United States Food & Drug Administration (USFDA) database. In India, this is the first reported case of mycobacterium infection associated with a face lifting thread procedure. It is important to note that the patients who undergo these procedures and the infections take place, then the thread cannot be fully extracted. There have been

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descriptions of pacemaker site infection [10], peritoneal dialysis infections [11], catheter-related bloodstream infections [12] caused by rapidly growing atypical mycobacteria. Although infection with this thread had not yet been described related to face-lifting, the surgeons should be aware of rapidly growing mycobacteria, exclusively when soft tissue manipulation is present. To minimize the risk of infection in the patients, strict sterilization during the procedure should be rigorously maintained.

Conclusion

Atypical mycobacterium infection associated with the face lifting definisse thread is very rare. However, with increasing the frequency of mycobacterium infections from the definisse threads at implantation sites, it would be worth considering these organisms as potential emerging pathogens in device-related infections. The article illustrates the importance of maintaining device sterility and clinical suspicion of mycobacterial infection when surgical implants are used, especially, when related to skin and soft tissues in order to avoid such type of incidence in future.

Patient Consent

The authors certify that they have obtained patient consent in a patient consent form. In the form, the patient has given her consent for the images and details used in the article for publication.

Conflicts of Interest

The authors declare no conflict of interest.

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